

WRITE UP
4,5TSDFSDDJSJDDDRSJDDHDHSJDDTHIDJSSSAL!!!!SLASL!



\$1=Rp.19000/19 juta lapangan pekerjaan duluan?
Boboiboy Kuase 3
sumbul

MISC.....	3
Sanity Check.....	3
Flag : GCW{welcome_to_gcw_finals_good_luck_have_fun_1337}.....	4
Lost In Translation.....	5
Flag :	
GCW{when_yh_bisa_ikutan_#kaburajadulu_jujur_muak_juga_lama2_gwe_aku_saya}	10
FORENSIC.....	11
Silent Execution.....	11
Flag : GCW{p0w3rsh3ll_3nc0d3d_b4s364_1n_RAM}.....	12
CRYPTOGRAPHY.....	13
Recovery.....	13
Flag : GCW{d373rm1n1571c_5rn6_l34k_v14_7r4c3_1d5_5h4m1r}.....	17

MISC

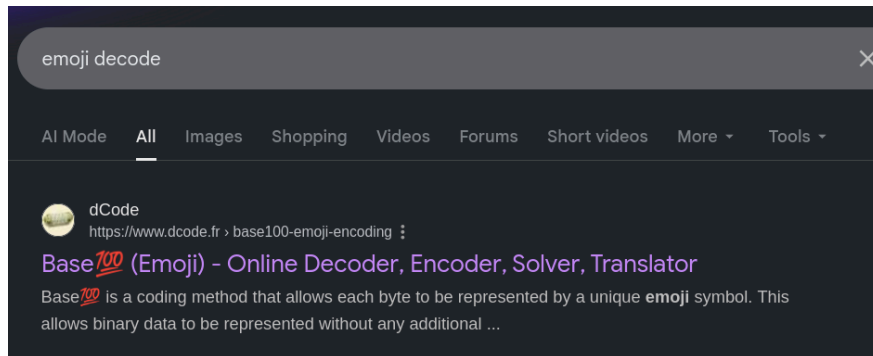
Sanity Check



Diberikan sebuah soal misc dengan deskripsi ibnu dengan emoji-nya

[👉🐧🍷👐👜🐪🛍️👣🎮🐾🍷👫🎮👉👍🍋🛍️🐾🐰🍋👑🐘🍷🐪👐🐘👑👩🎮👤🍋👥👕🐾🦊🐰👤🎮🐘👑👍🐰👤🛍️👉🐟🍋💋👐💋👩🍷🐪🐧🐎]

Langsung kita cari emoji decode dan kita mendapatkan web paling atas yaitu dcode



Lalu kita coba decode dengan web tersebut, dan menghasilkan sebuah strings base64



Lalu kita bisa decode dengan base64 dan didapatkan hasilnya yaitu sebuah flag

```
~
Lalu kita coba decode dengan web tersebut, dan menghasilkan sebuah string base64
> echo "RONXe3dlbGNvbWVfdG9fZ2N3X2ZpbmFsc19nb29kX2x1Y2tfaGF2ZV9mdW5fMTMzN30=" | base64 -d
GCW{welcome_to_gcw_finals_good_luck_have_fun_1337}
~
>
```

Flag : GCW{welcome_to_gcw_finals_good_luck_have_fun_1337}

MISC

Lost In Translation



Diberikan sebuah soal dengan deskripsi 'you're smart, you'll figure it out.' dan sebuah koneksi netcat **nc ctf.gcwug.com 1644**.

Pada saat saya mencoba koneksi tersebut

```
> ~
> nc ctf.gcwug.com 1644
Welcome to Lost in Translation.
Answer quickly or get cut off.
You'll get to the next stage if you answer correctly.
The answers must be exact.

[1/50] What is 893 + 558?
>
Too slow. Connection closed.

Ncat: Broken pipe.
> |
```

Bisa kita lihat bahwa koneksi ini memiliki 50 pertanyaan yang harus disubmit kurang lebih selama 2-3 detik. Di ke 50 pertanyaan itu, hanya ada 2 tipe soal yaitu

- [1/50] What is INTEGER + INTEGER?
- [1/50] Translate 'TEXT' to hex.

Disini saya membuat script automation untuk menjawab 2 tipe pertanyaan tersebut
Mulai dari tipe 'What is INTEGER + INTEGER?'

```
if "to hex" in data:
    target_match = re.search(r"'(.*)'", data)
    if target_match:
        text_to_convert = target_match.group(1)
        hex_val =
binascii.hexlify(text_to_convert.encode('utf-8')).decode('utf-8')
        s.sendall((hex_val + '\n').encode('utf-8'))
        print(f"[Sent hex]: {hex_val}")
```

Ia akan menjumlahkan secara otomatis dan send ke server, dan tipe soal yang 'Translate
'TEXT' to hex.'

```
match = re.search(r'What is (\d+)\s*\+\s*(\d+)\s*\?', data)
if match:
    num1, num2 = int(match.group(1)), int(match.group(2))
    answer = str(num1 + num2)
    s.sendall((answer + '\n').encode('utf-8'))
    print(f"[Sent integer]: {answer}")
    continue
```

Ia akan otomatis mengenkripsi TEXT yang ada didalam tanda petik dan otomatis send hasil
enkripsi tersebut ke server

Lalu tinggal kita buat looping dengan script berikut

solveNC.py

```
import socket
import re
import binascii

HOST = 'ctf.gcwug.com'
PORT = 1644

def solve_challenge():
    s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    s.connect((HOST, PORT))
```

```

for i in range(1, 55):
    try:
        data = s.recv(4096).decode('utf-8', errors='ignore')
        if not data:
            print("\n[Server disconnected]")
            break
        print(f"\n[Server]: {data.strip()}")

        match = re.search(r'What is (\d+)\s*\+\s*(\d+)\s*\?', data)
        if match:
            num1, num2 = int(match.group(1)), int(match.group(2))
            answer = str(num1 + num2)
            s.sendall((answer + '\n').encode('utf-8'))
            print(f"[Sent integer]: {answer}")
            continue

        if "to hex" in data:
            target_match = re.search(r"'(.*)'", data)
            if target_match:
                text_to_convert = target_match.group(1)
                hex_val =
binascii.hexlify(text_to_convert.encode('utf-8')).decode('utf-8')
                s.sendall((hex_val + '\n').encode('utf-8'))
                print(f"[Sent hex]: {hex_val}")

    except Exception as e:
        print(f"Error: {e}")
        break

if __name__ == "__main__":
    solve_challenge()

```

Ketika kita running hasilnya sebagai berikut

```
~/CaptureTheFlag/GCWUG/Final/lost
> python3 solveNC.py

[50/50] What is 930 + 689?
>
[Sent integer]: 1619

[Server]: OK

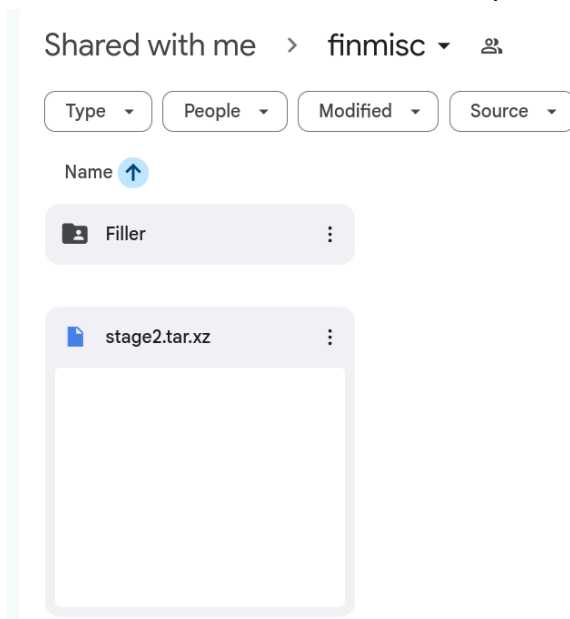
Nice. You survived stage 1.
Stage 2 unlocked: https://drive.google.com/drive/folders/1myUdM9fILGTdgZnnM74oggFHZ8Br70bf?usp=sharing
A new language awaits.
~/CaptureTheFlag/GCWUG/Final/lost
o >
```

Kita mendapatkan link google drive

 finmisc

bukan flag:v

Ketika kita buka Gdrive tersebut terdapat 1 folder dan 1 file zip



Singkat saja, folder filler isinya ngawimilton dan decoy flag. Jadi saya langsung fokus ke file [stage2.tar.xz](#), Di file ini kita mendapatkan 2 file yaitu [baca aku untuk petunjuk, kalo gamau gausah.txt] yang berisi sebuah hint

My Esoteric language gw rules aturan

```
Start with total = 0.
```

```
ADD X
```

```
total = (total + X) & 0xff
```

```
XOR X
```

```
total = total ^ X
```


OUT

```
print chr(total)
```

Every line is one instruction.

Dan file dialecto.txt yang berisi text XOR

Dari hint tersebut dapat disimpulkan bahwa untuk mendecrypt isi dari dialecto.txt kita hanya tinggal mengikuti hint tersebut

```
if cmd == "ADD":
    val = int(parts[1], 0)
    total = (total + val) & 0xFF

elif cmd == "XOR":
    val = int(parts[1], 0)
    total = total ^ val

elif cmd == "OUT":
    output.append(chr(total))
```

Lalu kita running script dengan rules esoteric diatas

XORDecrypt.py

```
import os

def run_bytecode_from_file(filename="dialecto.txt"):
    total = 0
    output = []

    with open(filename, "r", encoding="utf-8") as file:
        for line in file:
            line = line.strip()
            if not line or line.startswith("#"):
                continue

            parts = line.split()
            cmd = parts[0].upper()
```

```

        if cmd == "ADD":
            val = int(parts[1], 0)
            total = (total + val) & 0xFF

        elif cmd == "XOR":
            val = int(parts[1], 0)
            total = total ^ val

        elif cmd == "OUT":
            output.append(chr(total))

    return "".join(output)

if __name__ == "__main__":
    flag = run_bytecode_from_file("dialecto.txt")
    if flag is not None:
        print(flag)

```

Dan ketika running script tersebut kita berhasil mendecrypt isi dari file dialecto.txt dengan output

```

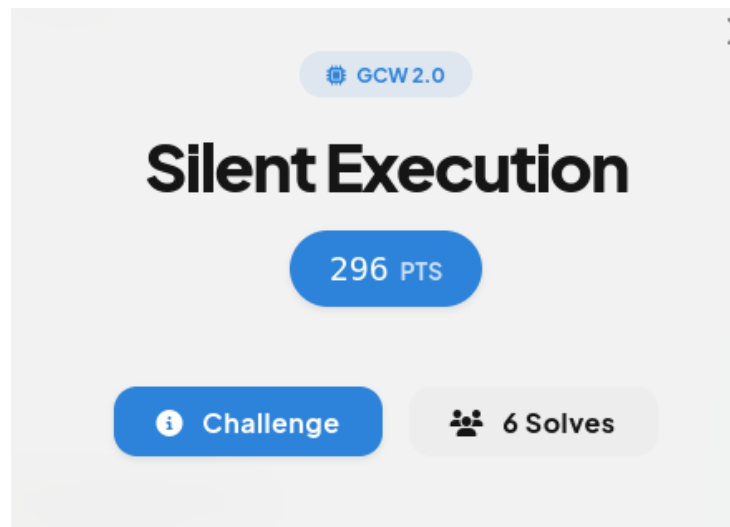
~/CaptureTheFlag/GCWUG/Final/lost
● > python3 XORsolve.py
Flag real beneran serious trust me bro oke? oke?? oke gas oke gas mari kita honeymoon ke paris: GCW{when_yh_bisa
_ikutan_#kaburajadulu_jujur_muak_juga_lama2_gwe_aku_saya}
~/CaptureTheFlag/GCWUG/Final/lost
○ >

```

Flag : GCW{when_yh_bisa_ikutan_#kaburajadulu_jujur_muak_juga_lama2_gwe_aku_saya}

FORENSIC

Silent Execution



Diberikan sebuah soal dengan deskripsi 'Sebuah workstation menjalankan sesuatu secara tersembunyi. Layar bersih, tapi memory tidak berbohong. Analisis dump RAM ini dan temukan apa yang dieksekusi.'

Untuk solving soal ini jujur kami hanya melakukan command

```
> strings chall.raw | grep -i "GCW{"
```

Lalu tunggu proses selesai sekitar 40 detik kita akan diberikan beberapa flag yang bisa saja decoy

```
~> .../GCWUG/Final/Silent
~/CaptureTheFlag/GCWUG/Final/Silent
> strings chall.raw | grep -i "GCW{"
$inner = '$f = "GCW{p0w3rsh3ll_3nc0d3d_b4s364_1n_RAM}"; Write-Output $f; Start-Sleep -Seconds 99999'
$inner = '$f = "GCW{p0w3rsh3ll_3nc0d3d_b4s364_1n_RAM}"; Write-Output $f; Start-Sleep -Seconds 99999'
$inner = '$f = "GCW{p0w3rsh3ll_3nc0d3d_b4s364_1n_RAM}"; Write-Output $f; Start-Sleep -Seconds 99999'
c:\windows\system32\windowpowershell\v1.0\powershell.exe -noprofile -windowstyle hidden -encodedcommand jabmacaapqagaciarwbdafeaw
bwadaadwazahiacwboadmababsaf8amwbuagmamabkadmazabfagianabzadmanga0af8amqbuaaf8augbbae0afqaiadsaiabxahiaaqb0agualqbpahuadabwahuadaaga
cqazga7acaaub0ageacgb0ac0auwbsaguazqbwacaaalqbtaguaywbvag4azabzacaa0qa5adkaoqa5aa==f = "gcw{p0w3rsh3ll_3nc0d3d_b4s364_1n_ram}"; wr
ite-output $f; start-sleep -seconds 99999
= "GCW{p0w3rsh3ll_3nc0d3d_b4s364_1n_RAM}"; Write-Output $f; Start-Sleep -Seconds 99999'
c:\windows\system32\windowpowershell\v1.0\powershell.exe -noprofile -windowstyle hidden -encodedcommand jabmacaapqagaciarwbdafeaw
bwadaadwazahiacwboadmababsaf8amwbuagmamabkadmazabfagianabzadmanga0af8amqbuaaf8augbbae0afqaiadsaiabxahiaaqb0agualqbpahuadabwahuadaaga
cqazga7acaaub0ageacgb0ac0auwbsaguazqbwacaaalqbtaguaywbvag4azabzacaa0qa5adkaoqa5aa==f = "gcw{p0w3rsh3ll_3nc0d3d_b4s364_1n_ram}"; wr
ite-output $f; start-sleep -seconds 99999
1GCW{f4ke_Flag}
1GCW{f4ke_Flag}
1GCW{f4ke_Flag}
c:\windows\system32\windowpowershell\v1.0\powershell.exe -noprofile -windowstyle hidden -encodedcommand jabmacaapqagaciarwbdafeaw
bwadaadwazahiacwboadmababsaf8amwbuagmamabkadmazabfagianabzadmanga0af8amqbuaaf8augbbae0afqaiadsaiabxahiaaqb0agualqbpahuadabwahuadaaga
cqazga7acaaub0ageacgb0ac0auwbsaguazqbwacaaalqbtaguaywbvag4azabzacaa0qa5adkaoqa5aa==f = "gcw{p0w3rsh3ll_3nc0d3d_b4s364_1n_ram}"; wr
ite-output $f; start-sleep -seconds 99999
1GCW{f4ke_Flag}
GCW{f4ke_Flag}
$inner = '$f = "GCW{p0w3rsh3ll_3nc0d3d_b4s364_1n_RAM}"; Write-Output $f; Start-Sleep -Seconds 99999'
/*gcw{
~/CaptureTheFlag/GCWUG/Final/Silent
```

46.577s

Saya coba satu satu dan gak ekspek salah satu dari hasil grep itu benar 🐱

Flag : GCW{p0w3rsh3ll_3nc0d3d_b4s364_1n_RAM}

CRYPTOGRAPHY Recovery



Diberikan soal ctf dengan 2 attachment yaitu [chall.py](#) dan output.txt yang isinya

```
session_id =  
71607302358858328329198675694601578129116266070302237101014533246904853748  
88  
trace_alpha =  
61440536045064128691097134029847638761687802416384796192237589001009960971  
122
```

```
trace_bravo =
37315296588380358445482742325820468750320938390564468768025417677942521718
991
trace_charlie =
20617629846758726153982233326380232750917323510355011808011337963071941558
209
share_bravo =
33256473601104795798385995777409145093413187116532670737846058558004098140
891
share_charlie =
11254366187588461932551641138811761290577310171356694450281477597806501946
1248
flag_nonce = '84dad38d62de66ab34952ca6177b68db'
flag_tag = '5125d12dd6ba9fc8227e9e023bc659e7'
flag_ct =
'ebb5931113b3beeb148a34cf6164f9669200eeddee5552e5e347de1c63fa744b560cfd6bd
9930a07cebaefc2d2410aff3a'
```

Tantangan (Challenge) ini melibatkan kriptografi berbasis fungsi acak linier (SRNG - *Simple Random Number Generator*) dan Polinomial Kuadrat. Untuk mendapatkan *flag*, kita perlu melakukan *reverse* terhadap generator untuk menemukan nilai *secret*, yang nantinya digunakan sebagai *key* AES-GCM.

Diketahui bahwa nilai trace saling terhubung melalui fungsi SRNG:

```
trace_alpha=(session_id . M+K) (mod p)
trace_brave=(trace_alpha . M+K) (mod p)
trace_charlie=(trace_brave . M+K) (mod p)
```

Kita dapat mencari nilai M dan K dengan selisih dari nilai trace tersebut:

```
diff_y = trace_charlie - trace_bravo
diff_x = trace_bravo - trace_alpha
```

```
M = (diff_y . diff_x-1) (mod p)
K = (trace_bravo . trace_alpha.M) (mod p)
```

Menghitung Nilai secret :

Setelah mendapatkan M dan K, kita bisa merekonstruksi nilai a_1 dan a_2 dari session_id:

$$A1 = (\text{session_id} \cdot M + K) \pmod{p}$$

$$A2 = (a1 \cdot M + K) \pmod{p}$$

Dengan memanfaatkan nilai $f(2)$ yaitu share_bravo, kita bisa membalikkan fungsi kuadrat untuk mencari secret:

$$\text{Share_bravo} + a1 \cdot 2 + a2 \cdot 4 \pmod{p}$$

$$\text{Secret} = \text{share_bravo} - 2a1 - 4a2 \pmod{p}$$

Dekripsi AES Gunakan nilai secret untuk membuat key SHA-256. Lakukan dekripsi AES-GCM menggunakan flag_nonce, flag_tag, dan flag_ct.

```
solver.py
import hashlib
from Crypto.Cipher import AES

session_id =
71607302358858328329198675694601578129116266070302237101014533246904853748
88
trace_alpha =
61440536045064128691097134029847638761687802416384796192237589001009960971
122
trace_bravo =
37315296588380358445482742325820468750320938390564468768025417677942521718
991
trace_charlie =
20617629846758726153982233326380232750917323510355011808011337963071941558
209
share_bravo =
33256473601104795798385995777409145093413187116532670737846058558004098140
891
share_charlie =
11254366187588461932551641138811761290577310171356694450281477597806501946
1248
flag_nonce_hex = '84dad38d62de66ab34952ca6177b68db'
flag_tag_hex = '5125d12dd6ba9fc8227e9e023bc659e7'
flag_ct_hex =
'ebb5931113b3beeb148a34cf6164f9669200eeddee5552e5e347de1c63fa744b560cfd6bd
9930a07cebaefc2d2410aff3a'
```

```

p = 0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFC2F

diff_y = (trace_charlie - trace_bravo) % p
diff_x = (trace_bravo - trace_alpha) % p

inv_diff_x = pow(diff_x, -1, p)

M = (diff_y * inv_diff_x) % p
K = (trace_bravo - (trace_alpha * M)) % p
a1 = (session_id * M + K) % p
a2 = (a1 * M + K) % p

secret = (share_bravo - (2 * a1) - (4 * a2)) % p

secret_verify = (share_charlie - (3 * a1) - (9 * a2)) % p

flag_key = hashlib.sha256(secret.to_bytes(32, 'big')).digest()
nonce = bytes.fromhex(flag_nonce_hex)
tag = bytes.fromhex(flag_tag_hex)
ct = bytes.fromhex(flag_ct_hex)

try:
    cipher = AES.new(flag_key, AES.MODE_GCM, nonce=nonce)
    flag = cipher.decrypt_and_verify(ct, tag)
    print(f"{flag.decode('utf-8', errors='ignore')}")
except Exception as e:
    print("failed")

```

Output


```
~/CaptureTheFlag/GCWUG/Final/recovery  
> python3 solv.py  
GCW{d373rm1n1571c_5rn6_l34k_v14_7r4c3_1d5_5h4m1r}  
~/CaptureTheFlag/GCWUG/Final/recovery  
> █
```

Flag : GCW{d373rm1n1571c_5rn6_l34k_v14_7r4c3_1d5_5h4m1r}